VRP and QAOA: Qubit Allocation Strategies

Drestanto Muhammad Dyasputro

1 What are VRP and QAOA?

- **VRP (Vehicle Routing Problem):** given a depot (start/end) and customers, find a short tour that visits each customer exactly once (and optionally multiple depots/vehicles in richer variants). It generalizes the TSP. A brute-force way to check all tours scales like trying all permutations ($\sim N!$ for N locations), which explodes quickly [1].
- **QAOA (Quantum Approximate Optimization Algorithm):** a hybrid quantum–classical method that encodes an objective (e.g., total distance) into a quantum "cost" Hamiltonian, alternates it with a mixing Hamiltonian, and classically tunes parameters (γ, β) to favor low-cost bitstrings. It is widely studied for combinatorial optimization [2, **?**, **?**].

$\Rightarrow$ We can *pose VRP as a QUBO/Ising model* and run QAOA to search for good (not necessarily optimal) routes.

2 Allocating qubits

(a) Theoretical minimum: logarithmic permutation encoding

- Idea: encode the *order of visits* directly as one of the $N!$ permutations. If a register of k qubits can represent 2^k distinct states, the information-theoretic lower bound to index all tours is

$$k_{\min} = \lceil \log_2(N!) \rceil.$$

- This is minimal because any injective encoding of $N!$ objects needs at least $\log_2(N!)$ bits/qubits. (Stirling's approximation shows $\log_2(N!) \sim N \log_2 N - 1.44N$.) The fact that a brute-force baseline enumerates $N!$ tours is standard for TSP/VRP families [1].

- Practical snag: turning a permutation index into a *local cost Hamiltonian* is hard. You end up doing a table lookup of tour lengths (or heavy classical post-processing), which undermines the benefits of in-circuit optimization.

(b) What people actually do

One-hot / edge (naive adjacency) encodings

- Common QAOA formulations use *one binary variable per directed edge* (or per city $\times$ time slot), yielding $\sim N^2$ variables/qubits for N locations. Cost = sum of chosen edge weights + penalties for degree/flow constraints. This is straightforward to Hamiltonian-encode and is widely used.
- Example reference that implements VRP with QAOA using an edge-based/QUBO form (scales with the number of edges, i.e., $\sim N(N-1)$): Azad *et al.* (2020) [3].
- Recent gate-based VRP/QAOA studies likewise discuss QUBO encodings and practical challenges at these qubit counts [4].
- For TSP specifically, many works and tutorials present *one-hot city-at-time* encodings ($O(N^2)$ qubits) [5, 6].

Binary "node-visit-time" (ranking/position) encodings

- More qubit-efficient is to give each customer a small *binary register* that stores their visit position (1...N). This uses about $N\lceil \log_2 N \rceil$ qubits and reconstructs the tour by sorting by the stored positions.
- Trade-off: the *cost/constraints get trickier* than one-hot (you must penalize duplicate positions and out-of-range values), but it remains practical within the QUBO/Ising framework and lets you tune penalties within QAOA.
- This style is closely related to general *Ising/QUBO mappings of NP problems* (see Lucas' survey) [7].

Encoding Method	**Qubit Count**	**Num of Qubit**	**Cost Function Simplicity**
Naive Matrix (NxN)	N^2	High	Simple
Logarithmic Permutation	$\lceil \log_2(n!) \rceil$	Low	Complex (impractical)
Node-Visit-Time	$n \cdot \lceil \log_2(N) \rceil$	Moderate	Moderate

3 Why not just use the theoretical minimum?

- Using only $\lceil \log_2(N!) \rceil$ qubits *packs the information* tightly, but the cost function becomes a *global lookup* over permutations rather than a *local polynomial* in qubits.
- Implementing that as a Hamiltonian either (i) requires precomputing all tour costs and effectively *brute-forcing* via post-processing, or (ii) produces highly non-local operators that are impractical on near-term hardware.
- In short: the minimal-qubit encoding *recovers the same combinatorial hardness* as enumerating all paths; one-hot and binary position encodings spend extra qubits to get *simple, local penalty terms* that QAOA can actually optimize [1, 7].

References

[1] Travelling salesman problem — wikipedia. `https://en.wikipedia.org/wiki/Travelling_salesman_problem`. Accessed: 2025-09-04.

[2] Edward Farhi, Jeffrey Goldstone, and Sam Gutmann. A quantum approximate optimization algorithm. *arXiv preprint*, 2014.

[3] Aram Azad et al. Solving vehicle routing problem using qaoa. *arXiv preprint*, 2020.

[4] Ryosuke Yasumura et al. Quantum-assisted vehicle routing: Realizing qaoa-based approach on gate hardware. *arXiv preprint*, 2025.

[5] Madushika Perera. Efficient encodings of the traveling salesman problem for quantum annealers and gate models. *arXiv preprint*, 2024.

[6] The Quantastic Journal. Traveling salesman problem using quantum computing. `https://medium.com/the-quantastic-journal/traveling-salesman-problem-using-quantum-computing-02ae6356544b`, 2019. Accessed: 2025-09-04.

[7] Andrew Lucas. Ising formulations of many np problems. *Frontiers in Physics*, 2:5, 2014.